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Kelas : S1IF-13-05

JAWABAN ASESMEN 3 SEARCHING & SORTING

1. Source Code jawaban :

```go
package main

import "fmt"

const NMAX = 1000000

type arrInt [NMAX]int

func SelectionSort(T *arrInt, n int) {
    for i := 0; i < n-1; i++ {
        idxMin := i
        for j := i + 1; j < n; j++ {
            if T[j] < T[idxMin] {
                idxMin = j
            }
        }
        T[i], T[idxMin] = T[idxMin], T[i]
    }
}

func median(T arrInt, n int) float64 {
    if n%2 == 1 {
        return float64(T[n/2])
    } else {
        mid1 := T[n/2-1]
        mid2 := T[n/2]
        return float64(mid1+mid2) / 2.0
    }
}

func main() {
    var A arrInt
```

```go
    var x int
    var n int = 0

    fmt.Println("Input data masukan :")
    fmt.Scan(&x)
    for x != -5313541 && n < NMAX {
        if x == 0 {
            if n > 0 {
                SelectionSort(&A, n)
                med := median(A, n)
                fmt.Println("Median :")
                fmt.Println(med)
            }
        } else {
            A[n] = x
            n++
        }
        fmt.Scan(&x)
    }
}
```

Screenshot Output :

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    PROBLEMS

PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal1> go run no1.go
Input data masukan :
7 23 11 0 5 19 2 29 3 13 17 0 -5313541
Median :
11
Median :
12
PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal1> go run no1.go
Input data masukan :
23 12 24 26 20 10 25 8 0 -5313541
Median :
21.5
PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal1> go run no1.go
Input data masukan :
19 21 34 67 32 1 0 45 32 44 76 0 99 89 0 -5313541
Median :
26.5
Median :
33
Median :
39
PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal1>
```

Penjelasan Singkat Cara Kerja Program :

Program ini bertugas membaca angka terus-menerus dan menghitung nilai tengah (median) setiap kali pengguna memasukkan angka 0.

2. Source Code Jawaban :

```go
package main

import "fmt"

const NMAX = 1001

type Pemain struct {
    namaDepan    string
    namaBelakang string
    gol          int
    assist       int
}

type arrPemain [NMAX]Pemain

func SelectionSort(T *arrPemain, n int) {
    for i := 0; i < n-1; i++ {
        maxIdx := i
        for j := i + 1; j < n; j++ {
            if T[j].gol > T[maxIdx].gol {
                maxIdx = j
            } else if T[j].gol == T[maxIdx].gol && T[j].assist >
T[maxIdx].assist {
                maxIdx = j
            }
        }
        temp := T[i]
        T[i] = T[maxIdx]
        T[maxIdx] = temp
    }
}

func main() {
    var n int
    var data arrPemain

    fmt.Println("Masukkan Data Input :")
    fmt.Scan(&n)

    for i := 0; i < n; i++ {
```

```go
        fmt.Scan(&data[i].namaDepan, &data[i].namaBelakang, &data[i].gol,
&data[i].assist)
    }

    SelectionSort(&data, n)

    fmt.Println("\nHasil Sorting :")
    for i := 0; i < n; i++ {
        fmt.Printf("%s %s %d %d\n", data[i].namaDepan,
data[i].namaBelakang, data[i].gol, data[i].assist)
    }
}
```

Screenshot Output :

```
OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   PROBLEMS

PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal2> go run no2.go
Masukkan Data Input :
9
Bruno Fernandes 7 3
Jamie Vardy 8 1
Dominic Calvert-Lewin 10 2
Son Heung-Min 9 2
Harry Kane 7 2
Ollie Watkins 6 1
Patrick Bamford 7 1
Callum Wilson 7 0
Mohamed Salah 8 2

Hasil Sorting :
Dominic Calvert-Lewin 10 2
Son Heung-Min 9 2
Mohamed Salah 8 2
Jamie Vardy 8 1
Bruno Fernandes 7 3
Harry Kane 7 2
Patrick Bamford 7 1
Callum Wilson 7 0
Ollie Watkins 6 1
PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal2> go run no2.go
Masukkan Data Input :
7
Bruno Fernandes 7 3
Dominic  Calvert-Lewin 7 12
Son Heung-Min 7 2
Harry Kane 5 2
Patrick Bamford 5 1
Callum Wilson 5 0
Mohamed Salah 3 2

Hasil Sorting :
Dominic Calvert-Lewin 7 12
Bruno Fernandes 7 3
Son Heung-Min 7 2
Harry Kane 5 2
Patrick Bamford 5 1
Callum Wilson 5 0
Mohamed Salah 3 2
PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal2>
```

Penjelasan Singkat Cara Kerja Program :
Program ini bertugas mengurutkan daftar pemain sepak bola berdasarkan jumlah gol terbanyak (Descending).

3. Source Code Jawaban :

```go
package main

import "fmt"

const NMAX = 1000000

type partai struct {
    nama  int
    suara int
}

type tabPartai [NMAX]partai

func posisi(t tabPartai, n int, nama int) int {
    for i := 0; i < n; i++ {
        if t[i].nama == nama {
            return i
        }
    }
    return -1
}

func main() {
    var p tabPartai
    var n int = 0
    var x int

    fmt.Println("Masukkan proses input suara :")

    for {
        fmt.Scan(&x)
        if x == -1 {
            break
        }

        idx := posisi(p, n, x)
        if idx != -1 {
```

```go
            p[idx].suara++
        } else {
            p[n].nama = x
            p[n].suara = 1
            n++
        }
    }

    for i := 1; i < n; i++ {
        temp := p[i]
        j := i - 1
        for j >= 0 && p[j].suara < temp.suara {
            p[j+1] = p[j]
            j--
        }
        p[j+1] = temp
    }

    fmt.Println("\nHasil Perhitungan suara :")
    for i := 0; i < n; i++ {
        if i > 0 {
            fmt.Print(" ")
        }
        fmt.Printf("%d(%d)", p[i].nama, p[i].suara)
    }
    fmt.Println()
}
```

Screenshot Output :

```
OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    PROBLEMS

PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal3> go run no3.go
Masukkan proses input suara :
5 8 8 6 6 6 8 6 7 5 6 8 7 7 6 6 6 5 8 7 6 7 8 8 7 8 7 5 8 7 8 5 7 6 6 6 5 6 8 6 6 7 7 8 7 5 6 8 8 5 6 6 5 5 7 8 5 8 7 7 8 5 7 7 5 6 7 6 5 8 8 7
 6 7 8 7 6 8 7 5 6 6 5 8 8 8 8 6 6 6 8 6 7 8 7 8 5 8 -1

Hasil Perhitungan suara :
8(29) 6(28) 7(24) 5(17)
PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal3> go run no3.go
Masukkan proses input suara :
14 10 13 13 14 10 11 13 13 12 15 11 10 -1

Hasil Perhitungan suara :
13(4) 10(3) 14(2) 11(2) 12(1) 15(1)
PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal3> go run no3.go
Masukkan proses input suara :
-1

Hasil Perhitungan suara :

PS C:\Users\useru\OneDrive\azzamy\praktikumsem2\1090582500091_Azzamy-Gusalim-Herfino\ASESMEN 3\Soal3>
```

Penjelasan Singkat Cara Kerja Program :
Program ini berfungsi sebagai mesin penghitung suara. Ia menghitung berapa kali nomor partai muncul, lalu menampilkannya dari suara terkecil ke terbesar.